

TASK 12 - NumPy Aggregations and Statistics

FULL COMPLETE ASSIGNMENT

1. Task Overview

Use NumPy functions to calculate sum, mean, median, minimum, maximum, standard deviation, and variance. The task also applies these functions to a numerical student-performance dataset.

2. Objective

Build basic numerical analysis skills using NumPy and understand how statistical summaries describe numerical data.

3. Tools

Python, NumPy, and Jupyter Notebook.

4. Aggregation Functions

Aggregation functions summarize numerical values into useful statistics. They help describe totals, averages, central values, extremes, and data spread.

5. Sum - np.sum()

Calculates the total of all values.

```
data = np.array([60, 70, 80, 90, 100])
np.sum(data)
# 400
```

6. Mean - np.mean()

Calculates the arithmetic average.

```
np.mean(data)
# 80.0
```

7. Median - np.median()

Returns the middle value after the values are ordered.

```
np.median(data)
# 80.0
```

8. Minimum and Maximum

np.min() returns the smallest value and np.max() returns the largest value.

```
np.min(data)
# 60

np.max(data)
# 100
```

9. Standard Deviation - np.std()

Measures how spread out values are around the mean. A smaller value indicates observations are closer to the mean.

```
np.std(data)
```

10. Variance - np.var()

Measures squared spread around the mean.

```
np.var(data)
```

11. Seven Required Statistical Functions

The task requires examples of at least seven aggregation/statistical functions.

```
np.sum(data)
np.mean(data)
np.median(data)
np.min(data)
np.max(data)
np.std(data)
np.var(data)
```

12. Student Performance Dataset

Each row represents a student and each column represents a subject.

```
marks = np.array([
    [78, 82, 91],
    [65, 74, 80],
    [88, 90, 95],
])
```

```
[55, 68, 72]  
])
```

13. Overall Dataset Statistics

Without an axis, the functions summarize all values in the 2D array.

```
np.sum(marks)  
np.mean(marks)  
np.median(marks)  
np.min(marks)  
np.max(marks)  
np.std(marks)  
np.var(marks)
```

14. axis=0 - Column-Wise Analysis

For this 2D dataset, axis=0 performs the operation down rows and returns one result for each column.

```
np.mean(marks, axis=0)  
np.min(marks, axis=0)  
np.max(marks, axis=0)
```

15. axis=1 - Row-Wise Analysis

axis=1 performs the operation across columns and returns one result for each row.

```
np.mean(marks, axis=1)  
np.sum(marks, axis=1)
```

16. axis=0 vs axis=1

axis=0 gives column-wise results for this matrix; axis=1 gives row-wise results.

Expression	Meaning
np.mean(marks)	Mean of all values
np.mean(marks, axis=0)	Mean of each column
np.mean(marks, axis=1)	Mean of each row
np.sum(marks, axis=1)	Total for each row

17. Practical Statistical Analysis

Column-wise statistics can compare subjects, while row-wise statistics can summarize individual students.

```
print("Column means:", np.mean(marks, axis=0))  
print("Row means:", np.mean(marks, axis=1))  
print("Column minimum:", np.min(marks, axis=0))  
print("Column maximum:", np.max(marks, axis=0))  
print("Row totals:", np.sum(marks, axis=1))
```

18. Short Interpretation

Mean describes average performance. Median identifies the middle value. Minimum and maximum show the lowest and highest observations. Standard deviation and variance describe the spread of values.

19. Common Mistakes

1. Confusing mean with median. 2. Confusing variance with standard deviation. 3. Mixing up axis=0 and axis=1. 4. Forgetting that a 2D array contains rows and columns. 5. Applying a statistic to the wrong data.

20. Interview Questions and Answers

What is the difference between mean and median? Mean is the arithmetic average; median is the middle value after ordering.

What does standard deviation measure? It describes the spread of values around the mean.

What does variance measure? It measures squared spread around the mean.

What does axis=0 do? For this 2D dataset, it produces column-wise results.

What does axis=1 do? For this 2D dataset, it produces row-wise results.

Why are aggregation functions useful? They summarize numerical data and support statistical analysis.

Name seven NumPy functions used in this task. sum, mean, median, min, max, std, and var.

21. Quick Revision

```
sum    -> total  
mean   -> average
```

```

median -> middle value
min     -> smallest value
max     -> largest value
std     -> spread around mean
var     -> squared spread
axis=0  -> column-wise result
axis=1  -> row-wise result

```

22. Complete Mini Program

```

import numpy as np

# TASK 12 - NumPy Aggregations and Statistics

# 1. One-dimensional numerical data
data = np.array([60, 70, 80, 90, 100])

print("SUM:", np.sum(data))
print("MEAN:", np.mean(data))
print("MEDIAN:", np.median(data))
print("MINIMUM:", np.min(data))
print("MAXIMUM:", np.max(data))
print("STANDARD DEVIATION:", np.std(data))
print("VARIANCE:", np.var(data))

# 2. Student performance dataset
marks = np.array([
    [78, 82, 91],
    [65, 74, 80],
    [88, 90, 95],
    [55, 68, 72]
])

print("\nOVERALL STATISTICS")
print("Sum:", np.sum(marks))
print("Mean:", np.mean(marks))
print("Median:", np.median(marks))
print("Minimum:", np.min(marks))
print("Maximum:", np.max(marks))
print("Standard deviation:", np.std(marks))
print("Variance:", np.var(marks))

# 3. Axis operations
print("\nCOLUMN-WISE RESULTS (axis=0)")
print("Mean:", np.mean(marks, axis=0))
print("Minimum:", np.min(marks, axis=0))
print("Maximum:", np.max(marks, axis=0))

print("\nROW-WISE RESULTS (axis=1)")
print("Mean:", np.mean(marks, axis=1))
print("Total:", np.sum(marks, axis=1))

# 4. Short interpretation
print("\nINTERPRETATION")
print("Mean shows average performance.")
print("Median shows the middle value.")
print("Minimum and maximum show the range.")
print("Standard deviation and variance describe data spread.")

```

23. Expected Learning Outcome

Calculate common statistics with NumPy, use axis=0 and axis=1, summarize numerical datasets, and interpret basic statistical results.

24. Top 5 Skills

1. NumPy
2. Statistical Analysis
3. Numerical Data Processing
4. Aggregation Functions
5. Python Data Analysis